Olist Marketplace Analysis

SQL Server case study | Delivery performance, customer reviews & retention

An end-to-end SQL analysis of the Brazilian E-Commerce Public Dataset by Olist. The project investigates where delivery performance falls short, how late delivery is associated with customer reviews, and whether first-order delivery timing helps explain repeat purchasing.

8.11%	54.03%	25 days	2.69%
Delivered orders late	Late reviewed orders rated 1–2 stars	Average carrier-to-customer time for late orders	Six-month retention

Business question

Where is the marketplace exposed to poor delivery performance, and how does it relate to customer experience and repeat purchasing?

Prepared by Kritika Sharma | SQL Server · SSMS

Data period: September 2016 to October 2018; partial months at both ends.

1. Dataset, Tools & Method

The analysis uses the Brazilian E-Commerce Public Dataset by Olist, covering about 99,000 orders. The work was performed in SQL Server using SSMS and nine related tables.

Tables used

orders	**order_items**	**customers**
payments	**reviews**	**products**
sellers	**geolocation**	**category_name_translation**

SQL techniques

CTEs, joins, subqueries, window functions (ROW_NUMBER and LAG), conditional aggregation, EXISTS, date functions, and percentile analysis.

Data checks and baseline

• Baseline: 99,441 orders, 96,096 unique customers, and 3,095 sellers.

• Eight orders marked delivered have no customer delivery date; six canceled orders have a populated delivery date. These exceptions were documented rather than silently excluded.

• 610 products have no category; categories were not guessed or imputed.

• Monthly volume is higher in 2018 than 2017, but 2016 and 2018 are partial years, so this was not presented as annual growth.

Definitions and analytical choices

An order is defined as late when its actual customer delivery date is after the estimated delivery date. The estimate is used because it represents the delivery promise shown to the customer. For retention, the cohort includes customers whose first delivered order was on or before 17 April 2018, giving each customer a full six-month observation window before the dataset ends.

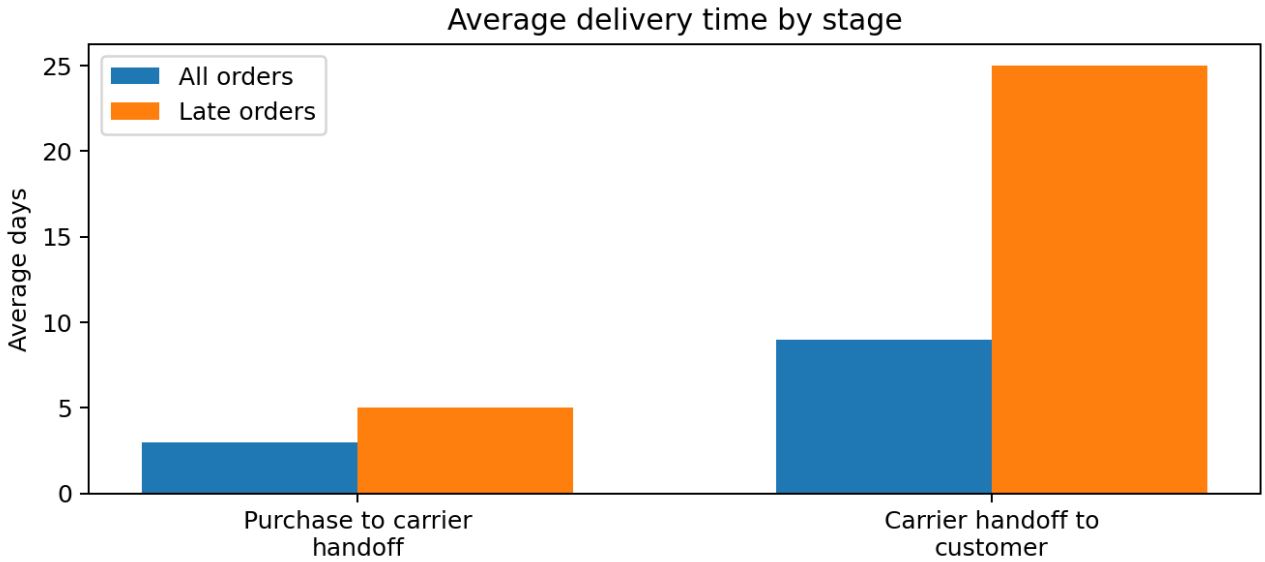
2. Delivery Performance

Delivery performance was assessed using the promised delivery date, the distribution of delivery time, and the size of the delay among late orders.

Metric	Result
Average delivery time	12 days
Median delivery time	10 days
75th / 90th / 95th percentile delivery time	16 / 23 / 29 days
Late orders	7,826 (8.11% of delivered orders)
Median delay among late orders	5 days
75th / 90th / 95th percentile of delay	11 / 21 / 29 days

Late orders carried R$1.35 million, equal to 8.76% of delivered-order payment value. This is payment value associated with late orders—an exposure measure, not evidence of lost revenue.

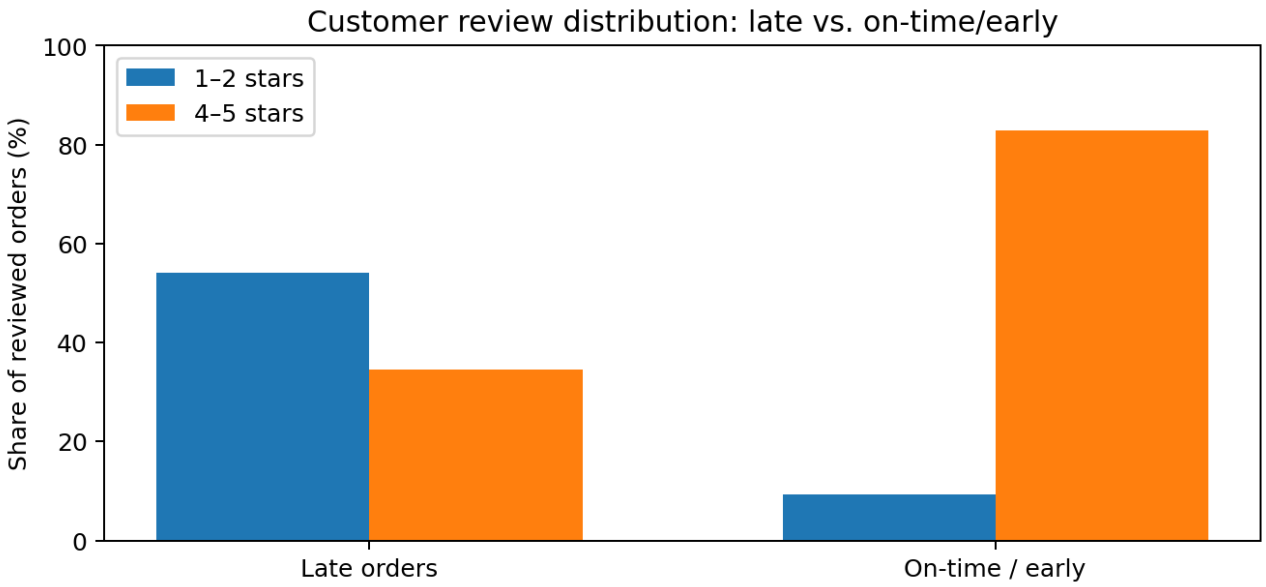
Where the time is spent



Late orders averaged 25 days from carrier handoff to customer, compared with 9 days for all orders. Purchase-to-carrier handoff time also increased, from 3 days overall to 5 days for late orders. This makes the post-handoff stage the first area to investigate, without attributing responsibility to a specific party.

3. Customer Experience & Reviews

The review distribution differs substantially between late orders and orders delivered on time or early.



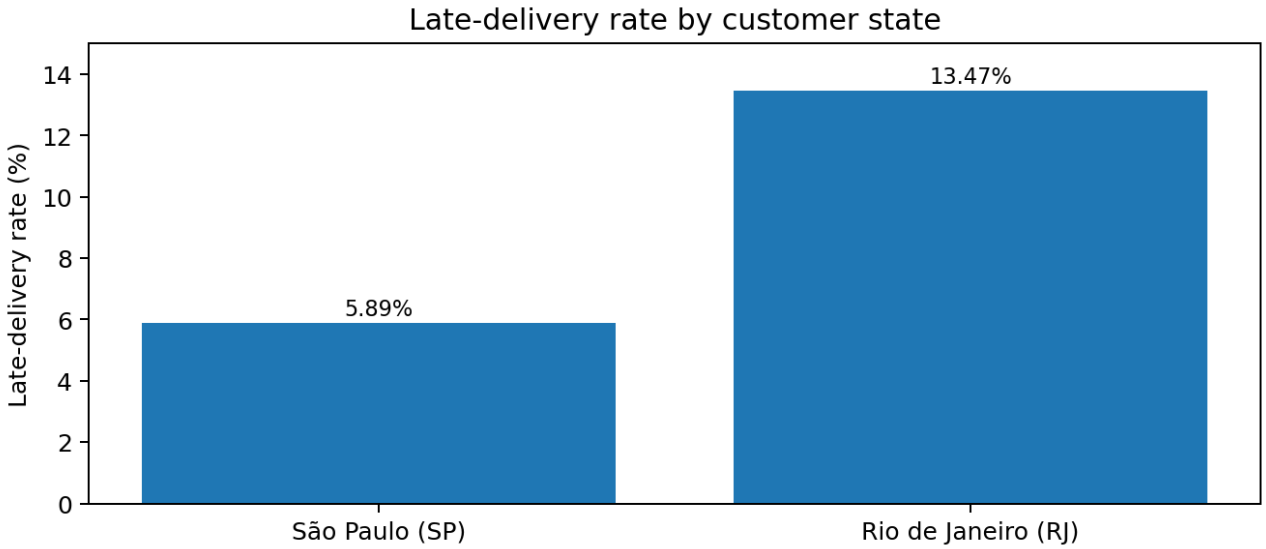
Delivery group	1–2 stars	4–5 stars
Late orders	54.03%	34.61%
On-time or early	9.23%	82.77%

For the slowest 5% of orders (delivery time over 29 days), average review score was 2.2, compared with 4.2 for all faster orders.

Interpretation: late delivery is strongly associated with poorer reviews in this dataset. The analysis is observational, so it does not prove that lateness alone caused the review differences; other order or customer factors may also matter.

4. Geographic Variation & Other Checks

Late-delivery rates vary by customer state.



Customer state	Late orders	Late rate
São Paulo (SP)	2,387	5.89%
Rio de Janeiro (RJ)	1,664	13.47%

SP has the largest count of late orders, largely reflecting its much larger order volume, while its late rate is lower than RJ's. Together, SP and RJ account for about 51.8% of all late orders.

Canceled and unavailable orders

Checks by month, product category, and seller did not reveal a strong concentration pattern. Higher rates appeared in small categories and low-volume sellers, where the number of observations was too small to support a reliable conclusion.

The state comparison supports prioritizing a closer review of RJ routes or the realism of estimated dates there. It does not by itself identify the operational cause of the difference.

5. Customer Retention

Only 2,997 of 96,096 customers (3.12%) ever ordered twice. This all-time measure is biased by the limited time available for customers who first purchased near the end of the dataset to return. A six-month cohort measure was therefore used for a fairer comparison.

First delivered order	Eligible customers	Returned	Six-month retention
Late	5,943	142	2.39%
On-time or early	59,974	1,631	2.72%
All	**65,917**	**1,773**	**2.69%**

Six-month retention is 2.69% overall. Customers whose first delivered order was late returned at 2.39%, compared with 2.72% for customers whose first order was on time or early. The difference is small, so the analysis does not support the conclusion that late first delivery explains low repeat purchasing.

Next analytical step

Treat retention as a separate business question and test other possible associations, such as first-purchase category or order value. These are proposed follow-up analyses, not findings from the current work.

6. Recommendations & Limitations

Recommendations

• **Track promised-date performance.** Use the late-delivery rate as a core KPI and report it alongside median and 90th-percentile delay, not just average delivery time.

• **Investigate the post-handoff stage first.** Late orders average 25 days from carrier handoff to customer versus 9 days overall; examine this stage before assigning responsibility.

• **Prioritize a closer look at Rio de Janeiro.** RJ's late rate is 13.47%, versus 5.89% in SP. Review routes and estimated-date accuracy, then monitor whether delivery and review outcomes change.

• **Investigate retention independently.** Six-month retention is 2.69%, and the first-delivery lateness comparison is only 2.39% versus 2.72%. Test other potential drivers.

Limitations

• The dataset covers a fixed historical period, with partial months at both ends.

• Review-score and delivery comparisons show association, not causation.

• Payment value associated with late orders is exposure, not lost revenue.

• Delivery-stage averages are rounded to whole days.

• Retention results apply to the defined cohort and may not generalize to other periods.

Conclusion

The analysis identifies a clear operational signal: late deliveries are associated with substantially poorer reviews, and much of the additional elapsed time occurs after carrier handoff. Performance also varies geographically, with RJ showing a higher late rate than SP. However, first-order lateness differs only slightly between retained and non-retained cohorts, so low retention requires separate investigation. These findings help prioritize the next questions without overstating causality.

Reproducibility: Import the nine Olist tables into SQL Server using the expected table names, then run the schema/setup script and main analysis script in the project repository.

Dataset: Brazilian E-Commerce Public Dataset by Olist on Kaggle — https://www.kaggle.com/datasets/olistbr/brazilian-ecommerce

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